

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

THE REVISION

We retain Neural Bellman Operators as the paper's subject and follow the report's NBO-centered path. The revision combines the original economic applications, one evaluation-improvement algorithm, an exact finite Bellman mechanism, and complete direct computational evidence. All 128 primary method executions, 32 independent mechanism assessments, classical scalar solves and deployments, and finite-observation comparisons are retained. Complete models and proofs remain current, and the unfavorable preceding comparisons and exact numerical records remain visible.

The direct comparison establishes an economically material NBO advantage over the declared neural HJB procedure in both dimensions after an independently proved deterministic transfer refinement. The comparison with cached Raw and DPO remains unresolved; the sufficient mechanism bound remains nonpositive, and the full-class absolute error account is wider than the direct-comparison margin. The replies identify these different conclusions precisely. A complete response does not imply that every requested stronger scientific conclusion has been established. The report is the advisory review stored in the repository, not an Econometric Society editorial decision.

B1: THE DISTINCTIVE NBO COMPONENT IS NOT SUPPORTED BY THE EXECUTED EVIDENCE

The current experiment identifies an economic contribution of the NBO evaluation-actor construction in the original dense capital economy. All 128 declared executions are complete, with no analytical fallback. In dimension fifty, the original simultaneous NBO-minus-neural-HJB interval is contained in $[0.001051, 0.001945]$, above the prespecified 10^{-4} materiality margin. NBO's schedule-relative method lower endpoints exceed 0.000943 and 0.000957 in dimensions ten and fifty. Tables II and III report the complete finite-distribution results, and Table IV reports all work.

The independently proved common-innovation transfer gives the sharper direct intervals in Table VII. NBO-minus-HJB is contained in $[0.000179, 0.000411]$ in di-

mension ten and $[0.001347, 0.001650]$ in dimension fifty; both establish superiority beyond the original 10^{-4} margin. The four refined Raw/DPO intervals remain unresolved. This reanalysis changes a deterministic allowance, not the observed paths, training, stopping, confidence allocation, or economic margin.

This finding concerns the declared common learned representation capacity, feasible action set, training schedules, stopping rule, and sixteen complete execution streams. The original ten-dimensional NBO–HJB interval remains inconclusive, as do the four NBO–Raw/DPO contrasts. We therefore distinguish an established benefit over this direct residual-fitting procedure from the separate empirical question of whether continuation regression improves on cached direct costate targets. The completed candidate-occupation assessment reports every mechanism term, but its sufficient welfare bound remains nonpositive; the independent method-payoff result does not substitute for a positive mechanism bound (our reply to B2).

Table I preserves all unfavorable earlier findings: 120 primary and 42 fixed-work direct intervals containing zero, twelve negative principal NBO-minus-Raw means, lower Raw work and larger Raw means in six fixed-work cells, and six critic diagnostics favoring Raw. The new result does not change those records or turn their inconclusive intervals into equivalence.

B2: THE COSTATE-TO-WELFARE MECHANISM IS NOT OPERATIONALLY CLOSED IN THE EXPERIMENT

The revision now evaluates every term of an exact finite Bellman mechanism. Section 6.3 defines the postdecision continuation target, the returned candidate’s occupation law, and its feasible action bridge. Proposition 7 gives the costate/action-change bound, Proposition 8 bounds the actual feasible actor’s gap, and Theorem 9 subtracts the holding deficit and quantitative payoff transfer. Appendix D supplies the complete proof and numerical accounts.

The independent assessment covers all 32 returned NBO policies. Each dimension uses 4,096 independent bridges, with two independent four-path antithetic continuation banks at every query. The costate cross-product removes reference-bank variance in expectation; the predictor’s error includes its training-grid discrepancy and the exact-state representation allowance. Global strong concavity is verified in

both dimensions, with a modulus lower bound greater than 1.056123. Sampling, clipping, arithmetic, holding, and diffusion-transfer allowances are all retained. Neither training-design MSE nor a nested-grid difference replaces the required candidate-occupation quantity.

Tables IX and X report the resulting evaluation and welfare accounts. Their continuous-gain lower bounds are approximately -0.006778 and -0.004800 . Thus the operationally evaluated sufficient bound does not certify positive welfare through critic accuracy at the executed precision. The critic-minus-Raw risk intervals also contain zero. These limitations remain next to the positive independent method-payoff comparison; that comparison is not substituted for the mechanism result.

B3: THE HIGH-DIMENSIONAL RESULT IS SAFE IMPROVEMENT, NOT AN ACCURATE SOLUTION OF THE CONTROL PROBLEM

The new study reports its absolute and relative accuracy objects together. Section 9.7 derives the full-adapted-class method regret bound from the original analytical anchor and each original method gain lower endpoint. Table VIII gives all eight outcomes. NBO's upper bounds are at most 0.025971 and 0.036843 in dimensions ten and fifty; the eight-method range is 0.025971–0.038358, approximately 260–384 times the 10^{-4} direct-comparison margin. These are valid explicit tolerances, not a claim of near-optimality at that margin.

The learned decisions nevertheless have a material measured economic consequence relative to an independently constructed neural HJB policy. The original direct comparison is materially positive in dimension fifty; the proved paired-transfer refinement establishes that result in both dimensions on the same observations. All sixteen NBO policies in dimension fifty also attain the actual online 0.0005 target with less measured complete work than DPO. These are direct economic and work findings for the stated algorithms, not a hidden replacement of the full adapted competitor class by the neural tube.

The scalar comparison adds an independent classical training and deployment study. Its full-box and matched-tube finite-grid arrays coincide at the saved solves, and Corollary S.16 bounds the exact finite-grid restriction loss by at most

1.788×10^{-11} on every initial-date node. The separate last state–time mesh changes remain approximately 1.72×10^{-4} – 3.25×10^{-4} . Classical deployed policies have higher observed payoffs than the two historical neural policies, and the NBO–DPO ranking reverses at $y = 1$. All these outcomes are retained. Neither the scalar algebraic account nor the nonpositive mechanism bound is presented as a tight high-dimensional continuous-time reference.

B4: THE STATISTICAL STATEMENTS ARE CONDITIONAL ON SELECTED FITTED OBJECTS, NOT STATEMENTS ABOUT THE METHODS’ TRAINING DISTRIBUTIONS

Section 9.3 defines a method as its initialization, training draws, optimization, validation, stopping, and failure rule. Its target is the uniform distribution over sixteen prospectively fixed complete execution streams. The experiment enumerated every stream for all four methods in both dimensions, giving 128 executions. All returned fitted policies; the fallback rule remains part of the algorithm but was not invoked.

Proposition 10 and its supplementary proof justify inference for this complete finite distribution, using 8,192 independent final paths per stream and 131,072 observations per method mean. Final-path uncertainty and the deterministic payoff-transfer account remain explicit. There is no missing-stream sampling layer, and no claim about an unrecorded distribution of future initializations or hardware environments. The original 238-event confirmation family is complete. The practical-equivalence margin remains 10^{-4} ; none of the four NBO–Raw/DPO intervals establishes equivalence. Existing historical policy-conditional intervals retain their original interpretation.

B5: THE WORK COMPARISON IS NOT YET A COMPLETE MATCHED-ACCURACY COMPARISON

Tables IV and V now give complete executed work: fitting, derivative and simulator work, action search, failed checks, checkpoint input and output, final confirmation, and peak memory. Raw retains and reuses its continuation labels. The comparison therefore does not charge it for a fictional repeated reconstruction of discarded targets.

At dimension fifty, all sixteen NBO, Raw, and DPO executions attained the online 0.0005 target, whereas HJB attained it in none. Mean complete times were approximately 357.80, 346.22, 596.94, and 1690.10 seconds, respectively. On the recorded environment NBO attained that common target with less work than DPO; Raw remained cheaper. HJB's unsuccessful executions are retained. The 900-second restricted-mean functional is a stated censoring convention, not a claim that every run consumed 900 seconds.

In dimension ten no method attained the online target. The subsequent independent confirmation certifies all sixteen NBO, Raw, and DPO policies above 0.0005, but it does not rewrite their earlier stopping decisions. The secondary 0.001 threshold has no positively certified stream. Complete checkpoint histories and stream endpoints remain in the supplement. Earlier retrospective prefix frontiers retain their retrospective interpretation. The process clock starts before fresh-worker launch and ends at durable final output. Dependency installation and artifact extraction precede this method-execution clock, as stated in Section 9.1.

B6: THE BENCHMARK SET IS TOO NARROW TO ESTABLISH A GENERAL NUMERICAL-METHOD CONTRIBUTION

Section 9.2 specifies four executed high-dimensional methods: NBO, cached anti-thetic Raw improvement, direct policy optimization, and direct neural HJB fitting. NBO and HJB use the same learned residual capacity and occupation information, with the correct different lifts for their distinct continuation and full-value targets. HJB uses two independent stochastic residual banks and pays for its gradient and action maximization at deployment. The declared update schedules and their costs remain visible; equal representation capacity is not described as identical optimization work.

The complete comparison establishes a material NBO–HJB payoff advantage in dimension fifty under the original transfer account and in both dimensions under the proved common-innovation refinement, together with the full work and stopping findings in our replies to B1 and B5. It does not establish superiority over Raw or DPO. The benchmark is an executed architecture-and-protocol comparison, not a theorem ranking all neural HJB solvers.

The independent scalar study also performs actual classical Howard training and measures deployment. It retains eight solves, three successively refined grids, a domain enlargement, both deployment meshes, all initial states, and all 110 paired comparisons. The classical policies' observed payoff exceeds the two frozen historical neural policies; the NBO–DPO ranking reverses at $y = 1$. Proposition S.15 and Corollary S.16 provide finite-grid algebraic and action-restriction bounds. These results supply an independent low-dimensional benchmark with its actual costs, without turning a scalar grid into a high-dimensional reference or counting an unexecuted BSDE comparison.

B7: THE MANUSCRIPT'S SCOPE AND ECONOMIC PAYOFF ARE NOT ALIGNED WITH THE RESULT ESTABLISHED BY R14

The paper remains Neural Bellman Operators. The main article presents one evaluation–improvement algorithm, general error accounts, the original capital economy, the operational mechanism, economic implementation, and direct evidence. Section 7 keeps the original applications visible; Appendix F retains their complete models, conditions, principal comparisons, and derivations. All preceding theorem, proposition, assumption, and equation labels remain in the current article or supplement.

Repeated iteration narratives and complete historical configuration tables remain in their exact archive, with every old label mapped by source line and Git blob. The unfavorable capital comparison and the worst whole-domain preference discrepancy remain current. Stylized initial-state populations and the proportional fee experiment remain explicitly stylized; their economic interpretation is not presented as a new calibration or as evidence of general computational superiority.

The capital model, feasible controller, exact paired-payoff identity, and policy-specific welfare theorem remain in the main article, followed by the new finite Bellman mechanism. Appendix C now groups the complete supporting anchor, diffusion-transfer, arithmetic, and Gaussian-tail accounts with their original statements and proofs. Moving those detailed constants makes the evaluation–action–payoff chain easier to follow without changing its assumptions or removing an economic application.

M1: CHOOSE THE PAPER'S CENTER

The paper retains Neural Bellman Operators as its subject, title, and economic program. One current evaluation–improvement algorithm connects the continuation object to a feasible actor and an economic payoff. The original recursive-utility, preference-adjustment, temporal-self, and game applications retain their distinct models and complete derivations.

The completed capital experiment establishes an economically material NBO advantage over the declared neural HJB procedure in both dimensions after the proved paired-transfer refinement, and lower measured work than DPO for the attained fifty-dimensional target. It also retains Raw's lower cost, the unresolved Raw/DPO payoff comparisons, and the nonpositive sufficient mechanism bounds. These outcomes identify the contribution and its scope. The common verifier supports the economic comparison; it is not substituted for the direct evidence about the NBO construction.

M2: DEFINE A METHOD-LEVEL ESTIMAND BEFORE ADDITIONAL EXPERIMENTS

The primary experiment was fixed at numerical source 9142f404; its complete evidence is committed at b0834744. The protocol identifies all sixteen complete streams, both dimensions, all four methods, the nine-profile initial-state population, three work levels, stopping and fallback rules, confirmation paths, and the 10^{-4} economic margin. The exact source and protocol SHA records are retained with the full evidence.

Section 9.3 defines the finite-method target before reporting results. All 128 prescribed executions and all 238 confirmation events are present, with no fallback or excluded stream. The 0.0005 online target and the 0.001 secondary final target have their distinct declared roles. No final payoff observation changes the stopped checkpoint, and an inconclusive direct interval is not relabelled equivalence.

The paired-transfer proof was developed after the original numerical-source freeze and inspection of the original transfer width. Appendix E gives the independent deterministic proof, and Table VII retains the original and refined intervals side by side. It introduces no new simulation, confidence event, stopping decision,

or fitted policy. We do not describe that later deterministic refinement as part of the prospectively frozen training protocol.

M3: MAKE THE CRITIC MECHANISM OPERATIONAL

The full chain is now computed on the same returned policies used by the primary experiment: exact finite-step continuation, candidate occupation, an independent uniform action bridge, costate error, scalar predicted advantage, feasible actor gap, holding deficit, and continuous-payoff transfer. The assessment source is b6b63511; all sixteen candidates in each dimension and all twelve registered simultaneous events are present. Tables K.I and K.II retain the individual statistical accounts, numerical cushions, and additional fresh-process work.

The descriptive critic/Raw risk means are approximately $4.499 \times 10^{-6}/3.018 \times 10^{-5}$ in dimension ten and $3.979 \times 10^{-5}/2.592 \times 10^{-5}$ in dimension fifty. They favor the critic in the first dimension and Raw in the second, but neither protected risk contrast determines the ranking. The product costate/action-change penalties are approximately 0.005630 and 0.004285; with the other retained terms they yield the nonpositive mechanism lower bounds reported in B2. No uncomputed error is set to zero, and the ten-dimensional descriptive reduction is not called simultaneous risk dominance or a causal explanation of the method-payoff result.

M4: REPORT COMPLETE ACCURACY-VERSUS-WORK FRONTIERS

The complete record now contains all three executed work checkpoints, every attempted certification, and the returned policy for every method and stream. In dimension fifty, NBO and Raw each attain the online 0.0005 target in three streams at checkpoint two and thirteen more at checkpoint three; DPO attains it in all sixteen at checkpoint three. HJB never attains it. In dimension ten all four methods complete their prescribed checks without attaining the online target. Table J.II preserves these actual histories.

Independent final certificates subsequently place all sixteen NBO, Raw, and DPO policies above 0.0005 in each dimension. This supplies a final economic-accuracy result, not a retrospectively fabricated earlier stopping time. The full-work table includes the work of unsuccessful checks and confirmation. The 0.001 final threshold

remains uncertified for every individual stream; no time-to-target result is assigned to it. These observations establish the stated measured-work advantage over DPO in the fifty-dimensional target experiment, while retaining Raw’s lower cost and every unattained outcome.

M5: ADD CONTRIBUTION-ISOLATING BASELINES

The comparison separates learned continuation pooling from cached direct continuation targets, direct policy differentiation, and direct residual fitting. NBO and Raw share the four-antithetic continuation construction and actor update/minibatch streams; Raw reuses the saved labels. NBO’s critic learning therefore has an observable incremental cost and is not credited with savings created by discarding its comparator’s information. HJB uses the same two-component learned representation capacity, while DPO differentiates the full declared finite-grid payoff.

The completed experiment reports all six direct NBO contrasts, all eight schedule-relative method results, the actual stopping histories, and complete work. The original fifty-dimensional NBO–HJB interval establishes a material payoff advantage; the proved common-innovation refinement establishes it in both dimensions. The Raw/DPO contrasts remain unresolved under both accounts. Classical scalar policies are independently fitted and deployed, and their observed advantage over the historical neural policies is retained. A separate actor is assessed through its measured deployment and overall procedure; the payoff comparison is not interpreted as a causal isolation of every architectural change.

M6: SEPARATE THE MODEL-SPECIFIC THEOREM FROM GENERIC CLAIMS

Sections 2–4 state the generic controlled-economy assumptions, the block-specific graph, comparison-based economic error bounds, and positive-weight approximation results. The capital sections separately use log felicity, consumption curvature, the dense production bound, the analytical schedule, and the quantitative paired-payoff identity. The finite Bellman mechanism explicitly states the continuation and action box to which it applies.

Section 7 and Appendix F give the recursive-utility domain, endogenous-preference normalization, temporal-self spike condition, and unilateral game crite-

1 rion. A theorem or experiment from one of these domains is not silently transferred 1
2 to another. 2

3 4 M7: CLARIFY THE ECONOMIC IMPLEMENTATION CLAIM 4

5 Section 9.10 reports a completed finite-observation experiment. A capital manager 5
6 receives measured log capital at the declared dates, computes consumption rates 6
7 with the saved network, and holds them until the next date. This rule needs neither 7
8 Brownian coordinates nor the intervening drift integral. The exact-history random- 8
9 ized representation remains separately stated with all its stronger assumptions. 9

10 Four protected parents were independently verified at 1,024 and 4,096 dates with 10
11 8,192 paths each, using the original fixed weights and volatility parameters. All 11
12 four parent lower gains are positive; six of twelve finite-observation lower gains 12
13 remain positive after the corresponding sensing allowance. At 4,096 dates and exact 13
14 measurements the lower gains are at least 0.001050 in dimension 10 and 0.001054 14
15 in dimension 50. An error of 0.0001 retains a lower gain of at least 0.000889 in 15
16 dimension 10 but is unresolved in dimension 50. Table XIII and supplementary 16
17 Tables M.V and M.VI report the complete allowances, ratios, and work. Large 17
18 allowances are not interpreted as evidence of a negative actual gain. 18
19

20 M8: REDUCE AND REORGANIZE THE PRESENTATION 20

21
22 The current reading copy has one algorithm and one system of notation. Main 22
23 results are followed by subject-based appendices rather than successive revision 23
24 manuscripts. All 230 labels in the reviewed source closure have a current or exact- 24
25 archive destination in the editorial map. The 24 original mathematical statements 25
26 and explicit proof environments pass a normalized full-body preservation check. 26
27 Complete application derivations and principal evidence remain current; duplicated 27
28 full configuration records and iteration narratives remain at the exact reviewed 28
29 commit identified in Appendix N. 29

30 The capital model, actual deployment, paired-payoff identity, policy-specific 30
31 welfare theorem, and new finite Bellman mechanism remain in the main article. 31
32 Supporting anchor, transfer, arithmetic, and tail constants and their complete proofs 32

are grouped in Appendix C. This rearrangement focuses the main narrative on evaluation, action, and economic payoff without a page-count deletion rule.

The publisher's class and configuration files are unchanged. Current supplementary results and equations have distinct sequential S. identifiers, while tables keep the publisher's appendix prefixes. The article uses the Econometrica class, author-year citations, numbered results, table legends, and cross-PDF references. Complete current evidence tables include every declared stream and checkpoint. The indexed archive makes each historical relocation inspectable.

M9: CLEAN UP METHOD LABELS AND EVIDENCE SEMANTICS

The four current identifiers are `nbo`, `raw_costate`, `direct_policy`, and `neural_hjb`. The original historical JSON files and hashes remain unchanged, including Raw records that carry the earlier `method: nbo` field and a `fixed_raw` experiment tag. Their interpretation is stated explicitly; they are not silently counted as newly trained NBO objects.

Current source, policy, stream, raw-array, and work identities accompany each of the 128 complete executions. Schedule-relative gains, direct trained-method contrasts, scalar finite-grid losses, full-class regret bounds, online target attainment, and independent final attainment have separate definitions and tables. In particular, the new material NBO-HJB result is a direct contrast under the declared finite distribution. Positive schedule gains do not establish Raw/DPO superiority, and their unresolved direct intervals do not establish equivalence. The original and independently refined transfer accounts retain distinct records and the same underlying observations.